

Homework Solutions

Calculus

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Problem P1

Suppose δ is a positive real number. How do you describe all real numbers x that are within δ of 0?

Solution. x within of 0: $|x| < \delta$, i.e. $-\delta < x < \delta$

Answer:

$$|x| < \delta, \text{ i.e. } -\delta < x < \delta$$

(a) The set of all x within δ of 0: $-\delta < x < \delta$ $[-\delta < x < \delta]$

(b) $|x| < \delta$ $|x| < \delta$

(c) $(-\delta, \delta)$ $(-\delta, \delta)$

Problem P2

Suppose a is a real number and δ is a positive real number. How do you describe all real numbers x that are within δ of a , but not equal to a :

Solution. x within of a , not equal to a : $0 < |x - a| < \delta$

Answer:

$$0 < |x - a| < \delta$$

(a) [Number line with open circles at endpoints, center marked] $[-\delta < x < \delta]$

(b) $a - \delta < x < a + \delta$, $x \neq a$ Or equivalently: $a - \delta < x < a$ or $a < x < a + \delta$

$$a - \delta < x < a + \delta, \quad x \neq a$$

(c) $0 < |x - a| < \delta$ $0 < |x - a| < \delta$

(d) $(a - \delta, a) \cup (a, a + \delta)$ $(a - \delta, a) \cup (a, a + \delta)$

Problem P3

Let $f(x) = x^2$.

Solution. Given: $f(x) = x^2$

Answer:

$$f(x) = x^2$$

(a) Need: $9.0 - 1 < x^2 < 9.0 + 1$ i.e. $8.0 < x^2 < 10.0$ Solving: $x \in (2.82842712474619, 3.162277660168)$
(positive x only) $(2.82842712474619, 3.16227766016838)$

(b) Need: $|x-3| < \delta \Rightarrow |x^2-9.0| < 1$ Boundaries: $x^2 = 10.0 \Rightarrow x = [-3.16227766016838, 3.16227766016838]$
 $x^2 = 8.0 \Rightarrow x = [-2.82842712474619, 2.82842712474619]$ Minimum distance from $x = 3$:
 $\delta = 0.16227766016838$ $\delta = 0.16227766016838$

(c) -

(d) Yes, this is true. For any $\varepsilon > 0$, we can find $\delta > 0$ such that $|x - a| < \delta \Rightarrow |x^2 - L| < \varepsilon$. This follows because $f(x)$ is continuous at $x = a$. Yes (continuity)

Problem P4

Graph $y = 2x - x^2$. For which points a is the tangent line to the curve at the point $(a, 2a - a^2)$ a horizontal line? How is the value of $2x - x^2$ at these points related to the values of $2x - x^2$ at other points?

Solution. Given: $y = -x^2 + 2x$

Derivative: $y' = 2 - 2x$

Horizontal tangent where $y' = 0$: $2 - 2x = 0$

Solutions: $x = [1]$

Points: $(1, 1)$

$f''(1) = -2 < 0$: this is a maximum.

The value 1 is the maximum of $-x^2 + 2x$.

Answer:

$(1, 1)$

Problem P5

The point $(2, 0)$ is on the curve $y = 2x - x^2$.

Solution. Given: $y = -x^2 + 2x$

Derivative: $y' = 2 - 2x$

At point $(2, 0)$:

Slope: $f'(2) = -2$

Tangent line: $y = 4 - 2x$

Answer:

$4 - 2x$

(a) Draw line through $(2, 0)$ with slope -2 Line with slope -2

(b) Slope at $x = 2$: -2 Tangent: $y - 0 = -2(x - 2)$ $y = 4 - 2x$ $y = 4 - 2x$

Problem P6

(a) Find $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{|x|} \right)$, $\lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right)$, and $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{|x|} \right)$, if they exist.

Solution. $\lim_{x \rightarrow 0^+} -\frac{1}{|x|} + \frac{1}{x} = 0$

$$\lim_{x \rightarrow 0^-} -\frac{1}{|x|} + \frac{1}{x} = -\infty$$

$$\lim_{x \rightarrow 0} -\frac{1}{|x|} + \frac{1}{x} = 0$$

Answer:

$$\boxed{0}$$

(b) Left limit: $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$ Right limit: $\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$ Since $-1 \neq 1$, the limit does not exist. $\boxed{\text{DNE}}$

Problem P7

Below are listed three limits. Match the best method to each limit and evaluate each limit.

Solution. See sub-problems for individual limit computations.

Answer:

$\boxed{\text{See sub-problems}}$

(a) $\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7}x^5 - 24} = -20$ $\boxed{-20}$

(b) $\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4} = 10$ $\boxed{10}$

(c) $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$ $\boxed{0}$

Problem P8

Using a calculator, try to guess $\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right)$. (Remember that n is in radians, not degrees!)

Solution. $\lim_{x \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right) = \pi$

Answer:

$$\boxed{\pi}$$

Problem P9

A wine barrel of height h and maximum radius R is formed by rotating the parabola $y = R - cx^2$ (for $-h/2 \leq x \leq h/2$) around the x -axis, where $c > 0$.

Solution. Simplified: h

Answer:

$$\boxed{h}$$

Problem P10

Draw a circle of radius 1. Inscribe a regular n -gon inside it.

∴

(a)

(b)

(c)

Auto-generated: 9/10 problems solved by SymPy.